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### Fitness score using heart rate recovery data

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#### Abstract

Embodiments provide functionality for determining fitness level of an individual using a fitness testing unit. In one such embodiment, an individual is instructed, via a graphical user interface (GUI) of a fitness testing unit, to perform a series of exercise activities. In turn, via the fitness testing unit, baseline heart rate, maximum heart rate, and recovery heart rate are captured while the individual is performing the series of exercise activities. Then, a fitness level of the individual is determined using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate.

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| <b>Inventors:</b>            | <b>Wiant; Matthew (Fairfield, CT), Iqbal; Mohammed (Arlington, VA)</b> |
| <b>Applicant:</b>            | <b>Myx Fitness, LLC (El Segundo, CA)</b>                               |
| <b>Family ID:</b>            | <b>70476548</b>                                                        |
| <b>Assignee:</b>             | <b>MYX Fitness, LLC (El Segundo, CA)</b>                               |
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*Primary Examiner:* Jackson; Gary

*Assistant Examiner:* Lukjan; Sebastian X

*Attorney, Agent or Firm:* Hamilton, Brook, Smith & Reynolds, P.C.

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## **Background/Summary**

RELATED APPLICATIONS (1) This application is the U.S. National Stage of International Application No. PCT/US2020/027738, filed on Apr. 10, 2020, published in English, which claims the benefit of U.S. Provisional Application No. 62/864,849, filed Jun. 21, 2019. The entire teachings of the above applications are incorporated herein by reference.

### **BACKGROUND**

(1) Fitness and, in particular, fitness instruction and tracking, have become increasingly prevalent. However, improved methods and systems for fitness instruction and tracking are needed.

### **SUMMARY**

(2) Embodiments provide improved fitness instruction and tracking through determination and use of a fitness score that is easy to understand and that facilitates tracking comparable fitness levels over long periods of time.

(3) An example embodiment determines a fitness level of an individual using a fitness testing unit. Such an embodiment, first, instructs an individual, via a graphical user interface (GUI) of the fitness testing unit, to perform a series of exercise activities. Next, via the fitness testing unit, a baseline heart rate, a maximum heart rate, and a recovery heart rate of the individual are captured while the individual is performing the series of instructed exercise activities. In turn, a fitness level of the individual is determined using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate.

(4) According to an embodiment, the series of exercise activities includes exercises of varying intensity. In one such embodiment, the series of exercise activities includes: a relaxation activity, a cardiovascular activity, and a recovery activity. In an embodiment, the baseline heart rate is captured while the individual is performing the relaxation activity, the maximum heart rate is captured while the individual is performing the cardiovascular activity, and the recovery heart rate is captured while the individual is performing the recovery activity. According to an embodiment, capturing the individual's heart rate includes continuously monitoring heart rate of the individual. Further, according to yet another embodiment, instructing the individual to perform the series of activities includes instructing the individual, via the GUI, to perform the cardiovascular activity

until it is determined that the monitored heart rate of the individual is at 80% of a recommended heart rate, wherein the recommended heart rate is based on the age of the individual.

(5) According to yet another embodiment, determining a fitness level of the individual using an age of the individual employs a factor indicating highest potential heart rate recovery given the age of the individual. Another embodiment includes obtaining the age of the individual by accessing a database storing a user profile associated with the individual. In this way, the age of the individual can be used in determining the fitness level of the individual.

(6) In embodiments, the fitness testing unit may be any such device capable of instructing an individual to perform exercise activities and capture heart rate of the individual. For example, the fitness testing unit may be implemented using a computing device including, memory, one or more processors, and one or more communication interfaces capable of performing the functionality described herein. Further, in an embodiment, the fitness testing unit comprises a bicycle or a treadmill.

(7) Another embodiment is directed to a fitness testing system that determines the fitness level of an individual. The system includes a processor and a memory with computer code instructions stored thereon that cause the fitness system to determine the fitness level of the individual. In an example embodiment, the system is configured to instruct an individual, via a GUI of the fitness testing system, to perform a series of exercise activities. While the individual is performing the series of exercise activities, the system captures a baseline heart rate, a maximum heart rate, and a recovery heart rate of the individual. Using the captured baseline heart rate, maximum heart rate, recovery heart rate, and an age of the individual, the system determines a fitness level of the individual.

(8) In an embodiment of the system, the series of exercise activities includes exercises of varying intensity. For instance, in an example embodiment, the series of exercise activities includes: a relaxation activity, a cardiovascular activity, and a recovery activity. In one such example embodiment of the system, in capturing the baseline heart rate, the maximum heart rate, and the recovery heart rate, the processor and the memory, with the computer code instructions, are configured to cause the fitness testing system to capture the baseline heart rate while the individual is performing the relaxation activity, capture the maximum heart rate while the individual is performing the cardiovascular activity, and capture the recovery heart rate while the individual is performing the recovery activity.

(9) Another embodiment of the system is configured to continuously monitor heart rate of the individual and instruct the individual, via the GUI, to perform the cardiovascular activity until it is determined that the monitored heart rate is at 80% of a recommended heart rate, wherein the recommended heart rate is based on the age of the individual. An embodiment of the system is configured to determine the fitness level of the individual using an age of the individual by employing a factor indicating highest potential heart rate recovery given the age of the individual. Yet another embodiment of the system is configured to obtain the age of the individual by accessing a database storing a user profile associated with the individual. Another example embodiment of the system is integrated into a bicycle or treadmill.

(10) An embodiment is directed to a computer program product for determining fitness level of an individual using a fitness testing unit. The computer program product comprises one or more non-transitory computer-readable storage devices and program instructions stored on at least one of the one or more storage devices. The program instructions, when loaded and executed by a processor associated with the fitness testing unit, cause the fitness unit to determine a fitness level of an individual as described herein. For instance, in an embodiment, the program instructions cause the fitness testing unit to instruct an individual, via a GUI of the fitness testing unit, to perform a series of exercise activities. Further, in such an embodiment, the program instructions cause the fitness testing unit to (1) capture a baseline heart rate, a maximum heart rate, and a recovery heart rate of

the individual while the individual is performing the series of exercise activities and (2) determine a fitness level of the individual using an age of the individual and the captured heart rates.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The foregoing will be apparent from the following more particular description of example embodiments, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments.
- (2) FIG. 1 is a flowchart of a method for determining the fitness level of an individual according to an embodiment.
- (3) FIG. 2 is a diagram of an embodiment of a system for determining the fitness level of an individual.
- (4) FIG. 3 is flowchart of a method for instructing an individual through a series of exercise activities and capturing heart rate of the individual while performing the activities according to an embodiment.
- (5) FIG. 4 is a simplified block diagram of a computing system for determining fitness level of an individual according to an embodiment.
- (6) FIG. 5 is a simplified diagram of a computer network environment in which an embodiment of the present invention may be implemented.

### DETAILED DESCRIPTION

- (7) A description of example embodiments follows.
- (8) The teachings of all patents, published applications, and references cited herein are incorporated by reference in their entirety.
- (9) Cardiac health, measured by the time it takes your heart rate to recover after a high intensity workout, is a meaningful measure of total fitness. Embodiments leverage heart rate recovery and determine a standardized fitness score assessment that is based on an individual's heart rate recovery and age. This standard measure allows a person to track improvement in cardiac health by tracking her fitness score over time. Embodiments determine a fitness score that accounts for age and starting heart rate level so that the fitness score is comparable over long periods of time.
- (10) FIG. 1 is a flowchart of method **100** for determining a fitness level of an individual using a fitness testing unit according to an embodiment. The method **100** begins at step **101** by instructing an individual, via a graphical user interface (GUI) of a fitness testing unit, to perform a series of exercise activities. According to an embodiment, the series of exercise activities includes exercises of increasing intensity with the goal of achieving a high percentage of maximum Heart rate to be used as the basis for HRR (Heart Rate recovery). The instructions in step **101** may be in any form that may be conveyed via the GUI. Further, an alternative embodiment of the method **100** may provide the instructions without the use of a GUI and may provide the instructions in any manner that can be perceived by an individual. Examples of such alternative methods for instructing the individual include mechanical instructions, e.g., vibration, and auditory instructions. Further, embodiments may use any combination of instruction modalities described herein.
- (11) In turn, at step **102**, a baseline heart rate of the individual, a maximum heart rate of the individual, and a recovery heart rate of the individual are captured while the individual is performing the series of exercises the individual was instructed to perform in step **101**. The heart rate of the individual may be captured at step **102** using any method of determining heart rate known in the art. For instance, in an embodiment, the individual wears a heart rate monitor that is communicatively coupled to the fitness testing unit and the heart rate monitor captures the individual's heart rate and communicates the individual's heart rate to the fitness testing unit via a

communication protocol, e.g., Bluetooth.

(12) To continue, at step **103**, a fitness level of the individual is determined using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate. According to an embodiment, the fitness level of the individual is determined by calculating a fitness score that is a function of the individual's age and captured heart rate data. According to an embodiment, the fitness score ranges from 0 to 100. Further, in an embodiment, the fitness level is determined at step **103** using the following equation:

$$(13) \text{ FitnessScore} = \frac{(100 \times (\text{MaxHR} - 1\text{MinRecoveryHR}))}{((\text{MaxHR} - \text{BaselineHR}) \times Z)}$$

where Baseline HR is the individual's base line heart rate measured while an individual is performing a relaxation activity, Max HR is the individual's maximum heart rate, 1 Min Recovery HR is the recovery heart rate measured after 1 minute of recovery, and Z is a heart rate recovery index based on age of the individual. Further details regarding calculating the fitness score and Z are described hereinbelow.

(14) In an embodiment, the fitness level of the individual is determined by a processor associated with the fitness testing unit executing computer code instructions stored in memory. In an embodiment, the fitness level of the individual is determined by the fitness testing unit. In an alternative embodiment, the fitness level of the individual is determined by a computing device that is located remotely in relation to the individual. In such an alternative embodiment, the captured heart rate data is communicated to the remotely located computing device and the remotely located computing device determines the fitness level. According to an embodiment, the device computing the fitness level of the individual obtains the age of the individual. In an embodiment, the age is obtained via user input at the time of determining the individual's fitness level or by accessing stored age data associated with the user. Another embodiment obtains the age of the individual by accessing a database storing a user profile associated with the individual. In this way, at step **103**, the age of the individual can be used in determining the fitness level of the individual. According to an embodiment, the age of the individual is used to select the Z value used to calculate the individual's fitness score. In another embodiment, gender and age of the individual are both used to select the Z value for calculating the fitness score.

(15) In an embodiment of the method **100**, the series of exercise activities the user is instructed to perform at step **101** includes: a relaxation activity, a cardiovascular activity, and a recovery activity. In such an embodiment, at step **102**, the baseline heart rate is captured while the individual is performing the relaxation activity, the maximum heart rate is captured while the individual is performing the cardiovascular activity, and the recovery heart rate is captured while the individual is performing the recovery activity. Further, according to yet another embodiment of the method **100**, instructing the individual to perform the series of activities at step **101** includes instructing the individual, via the GUI, to perform the cardiovascular activity until it is determined that the monitored heart rate of the individual is at 80% of a recommended heart rate. According to an embodiment, the recommended heart rate is based on the age of the individual. For instance, in one such example embodiment, maximum heart rate equals 220 minus age minus any adjustments which may be made for physical limitations.

(16) In an embodiment of the method **100**, capturing the individual's heart rate at step **102** includes continuously monitoring heart rate of the individual. Then, this continuously captured heart rate data is processed to identify the heart rate data needed, e.g., baseline heart rate, maximum heart rate, and recovery heart rate, to determine the individual's fitness level. According to yet another embodiment of the method **100**, determining a fitness level of the individual using an age of the individual employs a factor indicating highest potential heart rate recovery given the age of the individual. Such a factor may be pre-determined based upon previous heart rate data captured for a plurality of individuals. This data may be stored in memory that is accessible to a computing device determining the fitness level so that the factor may be employed in determining the individual's fitness level.

(17) Embodiments of the method **100** may be implemented using any computing device known in the art. For instance, the method **100** may be implemented on a computing device with instructions that cause a computer application to implement the method **100**. Further, the fitness testing unit may be any such device capable of instructing an individual, via a GUI, to perform exercise activities, and capture heart rate of the individual. For example, the fitness testing unit may be implemented using a computing device including, memory, one or more processors, and one or more communication interfaces capable of performing the functionality described herein. Further, in another embodiment, the fitness testing unit comprises an exercise device on which the exercises the individual is instructed to perform at step **101** can be performed. Example exercise devices include a bicycle, a treadmill, and a rowing machine, amongst other examples.

(18) Embodiments of the method **100** may perform a variety of functionalities using the fitness level, e.g., fitness score, determined at step **103**. One such embodiment displays the fitness level to the individual in the GUI. In another embodiment, the fitness level determined at step **103** is stored. In yet another embodiment, the fitness level determined at step **103** is used to modify the user experience in future exercise activities. For instance, in an embodiment, the fitness level determined at step **103** is used to create a personalized workout for the user. This personalized workout may be automatically determined based upon the fitness level or the fitness level may be provided to a fitness instructor, i.e., trainer, who creates a workout for the user based upon the determined fitness level. According to yet another embodiment, the fitness level determined at step **103** is aggregated with determined fitness levels for a plurality of other individuals. In this way, such an embodiment may track fitness levels for a population or specific group.

(19) The resulting fitness level determined at step **103** can also be tracked over time, which allows a user to assess his level of relative heart rate recovery and, therefore, track heart health over long periods of time. The fitness level, i.e., score, determined at step **103** allows for tracking fitness over a lifetime. Because the fitness level is determined at step **103** using the age of the individual, the fitness level measures relative health at any age. While heart rate recovery may change over time, the maximum fitness score, e.g., **100**, is attainable at any age, since the factors used to determine this maximum score adjust with age.

(20) The fitness level determined at step **103** can be used to automatically provide recommendations to users. For instance, the difficulty and intensity of workouts that users are instructed to perform can be automatically varied based on the user's fitness level. Likewise, fitness instructors can be recommended to a user based on the user's fitness level. In this way, a user can take a fitness class, e.g., view and remotely participate in a fitness class, that aligns with his or her fitness level. Further still, specific workouts and exercises can be recommended to a user based upon his fitness level in order to build the user's heart health. These aforementioned recommendations can all be done automatically, for example, through use of a database that stores instructor information, workout information, and recommended exercises and fitness levels that are appropriate for the various instructors, workouts, and exercises.

(21) FIG. 2 illustrates an environment **220** for determining the fitness level of an individual according to an embodiment. The environment **220** includes the fitness testing unit **221** that is configured to implement the embodiments described herein to determine a fitness level of the individual **224**. The fitness testing unit **221** comprises the bicycle **222** and computing device **223**. The environment **220** also includes the network **225** and server **226**.

(22) In an example implementation of the environment **220**, instructions **227** to perform a series of exercises are sent from the server **226** via the network **225** to the computing device **223**. The computing device **223** is configured to provide the instructions **227** in a graphical user interface on the display **229**. The individual **224** follows the instructions **227** and performs the series of exercises. The individual **224** wears a heart rate monitor **231** (that may be any heart rate monitor known in the art) that captures the individual's **224** heart rate data while the individual **224** is performing the series of exercises from the instructions **227**. The heart rate monitor **231**

communicates the heart rate data via the communication **230** (which may be a Bluetooth communication) to the computing device **223** of the fitness testing unit **221**. In an embodiment, the computing device **223** captures baseline heart rate, maximum heart rate, and recovery heart rate of the individual **224** while the individual **224** is performing the series of exercise activities. The computing device **223** determines a fitness level of the individual, e.g., computes a fitness score using the heart rate data obtained via the communication **230**. Alternatively, the computing device **223** can send the heart rate data **228** via the network **225** to the server **226** and the server **226** or another associated computing device can determine the fitness level of the individual **224**. Further, in an embodiment where the computing device **223** determines the individual's **224** fitness level, the computing device **223** may send the determined fitness level to the server **226**. Moreover, in embodiments of the environment **220**, the computing device **223** and/or server **226** may store the determined fitness level, e.g., fitness score of the individual. Further, in the environment **220**, multiple fitness levels of the individual **224** may be determined over time and stored by the computing device **223** and/or server **226** so that the individual **224** may measure and track fitness progress over time.

(23) As part of determining a fitness level of an individual, embodiments determine a fitness score. In an embodiment, a standardized heart rate recovery workout, such as the workout the user is instructed to perform in the method **330** is used to determine an individual's fitness score. FIG. **3** illustrates an example method **330** for instructing an individual to perform a series of exercise activities and capture the individual's heart rate data while performing the activities. In the method **330**, a participant is taken through a pre-set workout and the participant's heart rate is measured at pre-established times to determine baseline heart rate, maximum heart rate, and recovery heart rate.

(24) The method **330** begins at step **331** by instructing an individual to perform a relaxation activity. While the individual is performing the relaxation activity, baseline heart rate is captured at step **332**. At step **333**, the individual is instructed to perform a cardiovascular activity and at step **334** the individual's heart rate while performing the cardiovascular activity is captured. To continue, at step **335** the individual is instructed to perform a recovery activity, such as deep breathing, and at step **336** recovery heart rate is measured.

(25) According to an embodiment, at step **333**, the individual is instructed to perform activities that increase the individual's heart rate to approximately 80% of the recommended maximum heart rate for the individual's age. In an embodiment, the maximum heart rate for any given age is calculated with the accepted industry standard:

Maximum Heart Rate=220–Age of Individual–any adjustments which may be made for physical limitations

The cardiovascular activities the individual is instructed to perform at step **333** may be any such activities known in the art, e.g., running and/or cycling. Further, the instructions may be tailored and varied over time so that the user's heart rate reaches a desired level, e.g., 80% of recommended maximum heart rate.

(26) In an embodiment of the method **330**, baseline, maximum, and recovery heart rates may be captured at pre-determined times or, heart rate data may be captured and processed to determine baseline, maximum, and recovery heart rate. For example, heart rate data may be captured continuously while the individual is being instructed to, and is performing the relaxation activity. This continuously captured heart rate data may be processed to identify a minimum heart rate for the individual and the minimum heart rate would be the individual's baseline heart rate. Similar procedures may be employed for capturing maximum heart rate at step **334** and recovery heart rate at step **336**. In an embodiment of the method **330**, recovery heart rate may be captured at step **336** multiple times and at pre-determined times. For instance, in an example embodiment, the recovery heart rate is captured one minute and two minutes after starting to instruct the individual to perform the relaxation activity at step **335**. Further, in another embodiment, the individual is instructed to perform the relaxation activity at step **335** upon reaching the maximum heart rate, e.g., 80% of

recommended maximum heart rate, while performing the instructed cardiovascular activity. In such an embodiment, recovery heart rate may be captured at step **336** one minute and two minutes after the maximum heart rate (80% of recommended max) is reached.

(27) The method **330** may be performed over a pre-determined period of time. For example, the user may be instructed at step **331** to perform 2 minutes of relaxation, e.g., mindfulness, being at rest, etc., activities. During these 2 minutes, heart rate data is captured **332** and the lowest point is taken as the baseline heart rate. Then, at step **333**, the user is instructed to perform a cardiovascular activity for 15 to 25 minutes to ramp up heart rate and then, the highest heart rate captured **334** during that time period is the maximum heart rate. In turn, at step **335**, the user is instructed to perform a recovery/cool down protocol, e.g., deep breathing, and heart rate is captured **336** after 1 minute of cooling down and after 2 minutes of cooling down.

(28) As part of determining a fitness level of an individual, embodiments may determine a fitness score. In an embodiment, a fitness score is determined using the following equation, where the best possible score is 100 and the worst possible score is 0:

$$(29) \text{FitnessScore} = \frac{(100 \times (\text{MaxHR} - 1\text{MinRecoveryHR}))}{((\text{MaxHR} - \text{BaselineHR}) \times Z)}$$

where Baseline HR is the individual's base line heart rate measured while an individual is performing a relaxation activity, e.g., the heart rate measured at step **332** of the method **330**, Max HR is the individual's maximum heart rate, e.g., the heart rate measured at step **334** of the method **330**, 1 Min Recovery HR is the recovery heart rate measured after 1 minute of recovery, e.g., the heart rate measured at step **336** of the method **330**, and Z is a heart rate recovery index based on the age of the individual. It is noted that while the above equation utilizes 1 Min Recovery HR (the recovery heart rate measured after 1 minute of recovery), embodiments are not so limited and may use a recovery heart rate measured at various points in time, e.g., 1 minute, 2 minutes, 5 minutes etc. In an embodiment, the heart rate recovery data used is based on past measurements, i.e., if 1 minute heart rate recovery was used in the past, 1 minute heart rate recovery is used again so that a user can track his fitness level over time. Further, in an embodiment, the heart rate recovery data used is selected to correspond to the Z (heart rate recovery index) that is used in calculating the fitness score. As described hereinbelow, Z is based on fitness level scores from a plurality of individuals. If Z is calculated using 2 minute heart rate recovery data for the plurality of individuals, then, 2 minute heart rate recovery data is used when using said Z score. Similarly, if Z is calculated using 1 minute heart rate recovery data, then, 1 minute heart rate recovery data is used when using said Z score to calculate a fitness score. In this way, in such an embodiment, the heart rate recovery data and Z are chosen to correspond and, Z is also selected to correspond with the individual's age. Further, in embodiments, Z may also be chosen to correspond with the individual's gender. In other words, Z may be calculated for particular genders, e.g., there is a Z value for 30 year old females and a different Z value for 30 year old males. In an embodiment, Z represents the highest potential recovery of heartrate after 2 minutes for people of that age. According to an embodiment, Z indicates the best possible fitness level an individual may attain at her age. In this way, embodiments account for age when determining fitness level. In embodiments, Z is a number derived from actual heart rate recovery data from a plurality of individuals and Z represents the maximum potential percentage recovery from a maximum heart rate to the baseline heart rate for someone of a certain age. Further, Z may also be specific to gender. In other words, in such an embodiment, Z represents the maximum potential percentage recovery from a maximum heart rate to the baseline heart rate for someone of a certain age and gender. Further details and examples for determining Z are described hereinbelow.

(30) To illustrate calculating a fitness score, consider the example, where baseline heart rate is 75 beats per minute (BPM), 80% maximum heart rate is 180 BPM, recovery heart rate after 1 minute is 152 BPM, recovery heart rate is 125 BPM after 2 minutes, and Z (based on the user's age of 45) is 45%. Given these values, the individual's fitness score is:

(31) Example1:  $\text{FitnessScore} = 59.3 = \frac{(100 \times (180 - 152))}{((180 - 75) \times .45)}$

(32) Another example participant has a low fitness level. In this example, baseline heart rate is 75 beats per minute (BPM), 80% maximum heart rate is 180 BPM, recovery heart rate after 1 minute is 173 BPM, recovery heart rate is 164 BPM after 2 minutes, and Z (based on the user's age of 45) is 45%. Given these values, the individual's fitness score is:

(33) Example2:  $\text{FitnessScore} = 14.8 = \frac{(100 \times (180 - 173))}{((180 - 75) \times .45)}$

(34) In another example, the user is 65 years old, baseline heart rate is 75 beats per minute (BPM), 80% maximum heart rate is 160 BPM, recovery heart rate after 1 minute is 132 BPM, recovery heart rate is 105 BPM after 2 minutes, and Z (based on the user's age of 65) is 48%. Given these values, the individual's fitness score is:

(35) Example3:  $\text{FitnessScore} = 68.6 = \frac{(100 \times (160 - 132))}{((160 - 75) \times .48)}$

(36) In yet another example, the user is 65 years old, baseline heart rate is 75 beats per minute (BPM), 80% maximum heart rate is 160 BPM, recovery heart rate after 1 minute is 154 BPM, recovery heart rate is 145 BPM after 2 minutes, and Z (based on the user's age of 65) is 48%. Given these values, the individual's fitness score is:

(37) Example4:  $\text{FitnessScore} = 14.7 = \frac{(100 \times (160 - 154))}{((160 - 75) \times .48)}$

(38) The above examples illustrate the power of the determined fitness score and how the fitness score varies with age. The users in Example 1 and Example 3 have the same heart rate recovery, 28 beats per minute in 1 minute, however, the user in Example 1 is less fit (fitness score 59.3) than the user in Example 3 (fitness score 68.6) because the user in Example 1 is younger and should have a greater drop in heart rate from his max. The users in Examples 2 and 4 have similar fitness scores, but the user in Example 4 has a much lower max heart rate and a slower heart rate recovery (as expected for his age) and, thus, the users in Examples 2 and 4 have comparable fitness levels despite their differing scores for max heart rate and recovery heart rate.

(39) In an embodiment, fitness scores can be used to track fitness level over time. When someone first starts on a fitness plan, his heart rate recovers more slowly after exertion. Initially, a person's heart rate may drop only 10-15 BPM after 1 minute, or 20-25 BPM after 2 minutes. As the person progresses and becomes more fit, he can expect a heart rate drop of 20-30 BPM after 1 minute and 40-60 BPM or more after 2 minutes depending on age. These improvements can be measured and tracked overtime by repeating the methods described herein.

(40) According to an embodiment, Z is based on historical heart rate data collected from a plurality of individuals. For example, in embodiments, the historical heart rate data may be collected from a plurality of individuals while performing methods described herein, e.g., the method **100** or the historical heart data may be existing data, such as data from a medical study, amongst other examples. In an embodiment, Z is calculated using the following equation:

(41)  $Z = \frac{(\text{AverageHRR} + (3 \times \text{STD}))}{(\text{AveragePeakHR} - \text{AverageRestingHR})}$

where Average HRR is the average heart rate recovery amongst individuals of a particular age or age cohort, STD is the standard deviation of the heart rate recovery data of the individuals, Average Peak HR is the average peak heart rate measured while individuals of the particular age (or age cohort) are performing cardiovascular activity, and Average Resting HR is the average resting heart rate of individuals of that particular age (or age cohort). According to embodiment, Z is calculated for various age cohorts, e.g., 30 to 39 years old, 40 to 49 years old, 50 to 59 years old, etc. In an alternative embodiment, a Z value is calculated for each individual age, e.g., there is a Z for age 40, a Z for age 41, etc. In embodiments, Z may also be calculated for particular genders as well as ages. In such embodiments, the same calculation of Z as described above is used however, the heart rate recovery data used for calculating Z is all collected from male or female participants depending on whether a male Z value or female Z value is being calculated. When calculating a fitness score as described herein, a Z value is selected that corresponds to the age and/or gender

and/or time period of recovery of the individual for which the fitness score is being calculated.

(42) Further, it is noted that embodiments may modify the Z value calculation so as to modify the range of possible fitness scores. For instance, the above equation calculates a Z score that is based upon three standard deviations (the equation component  $3 \times \text{STD}$ ), however, embodiments may calculate a Z value based upon a different number of standard deviations. For example, Z may be calculated based upon two standard deviations. In such an example, Z is given by the following equation:

$$(43) Z = \frac{(\text{AverageHRR} + (2 \times \text{STD}))}{(\text{AveragePeakHR} - \text{AverageRestingHR})}$$

(44) To illustrate calculating Z, consider the example where heart rate data was collected for 1000 people between the ages of 30 to 39, and from this data, the average heart rate recovery 1 minute after stopping the cardiovascular exercise is 22.9, the standard deviation of the heart rate recovery data 1 minute after stopping the cardiovascular exercise is 8.34, the average peak heart rate is 182.4, and the average resting heart rate is 76.1. Given these values, Z for the 30 to 39 cohort is:

$$(45) Z = 45.1\% = \frac{(22.9 + (3 \times 8.34))}{(182.4 - 76.1)}$$

(46) FIG. 4 is a simplified block diagram of a computer-based system **440** that may be used to determine fitness level of an individual according to any variety of the embodiments of the present invention described herein. The system **440** comprises a bus **443**. The bus **443** serves as an interconnect between the various components of the system **440**. Connected to the bus **443** is an input/output device interface **446** for connecting various input and output devices such as a keyboard, mouse, display, speakers, etc. to the system **440**. A central processing unit (CPU) **442** is connected to the bus **443** and provides for the execution of computer instructions. Memory **445** provides volatile storage for data used for carrying out computer instructions. Storage **444** provides non-volatile storage for software instructions, such as an operating system (not shown). The system **440** also comprises a network interface **441** for connecting to any variety of networks known in the art, including wide area networks (WANs) and local area networks (LANs).

(47) It should be understood that the example embodiments described herein may be implemented in many different ways. In some instances, the various methods and machines described herein may each be implemented by a physical, virtual, or hybrid general purpose computer, such as the computer system **440**, or a computer network environment such as the computer environment **550**, described herein below in relation to FIG. 5. The computer system **440** may be transformed into the machines that execute the methods described herein, for example, by loading software instructions into either memory **445** or non-volatile storage **444** for execution by the CPU **442**. One of ordinary skill in the art should further understand that the system **440** and its various components may be configured to carry out any embodiments or combination of embodiments of the present invention described herein. Further, the system **440** may implement the various embodiments described herein utilizing any combination of hardware, software, and firmware modules operatively coupled, internally, or externally, to the system **440**.

(48) FIG. 5 illustrates a computer network environment **550** in which an embodiment of the present invention may be implemented. In the computer network environment **550**, the server **551** is linked through the communications network **552** to the clients **553a-n**. The environment **550** may be used to allow the clients **553a-n**, alone or in combination with the server **551**, to execute any of the methods described herein.

(49) Embodiments or aspects thereof may be implemented in the form of hardware, firmware, or software. If implemented in software, the software may be stored on any non-transient computer readable medium that is configured to enable a processor to load the software or subsets of instructions thereof. The processor then executes the instructions and is configured to operate or cause an apparatus to operate in a manner as described herein.

(50) Further, firmware, software, routines, or instructions may be described herein as performing certain actions and/or functions of the data processors. However, it should be appreciated that such

descriptions contained herein are merely for convenience and that such actions in fact result from computing devices, processors, controllers, or other devices executing the firmware, software, routines, instructions, etc.

(51) It should be understood that the flow diagrams, block diagrams, and network diagrams may include more or fewer elements, be arranged differently, or be represented differently. But it further should be understood that certain implementations may dictate the block and network diagrams and the number of block and network diagrams illustrating the execution of the embodiments be implemented in a particular way.

(52) Accordingly, further embodiments may also be implemented in a variety of computer architectures, physical, virtual, cloud computers, and/or some combination thereof, and thus, the data processors described herein are intended for purposes of illustration only and not as a limitation of the embodiments.

(53) While example embodiments have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the embodiments encompassed by the appended claims.

## Claims

1. A method for determining a fitness level of an individual using a fitness testing unit, the method comprising: instructing an individual, via a graphical user interface (GUI) of a fitness testing unit, to perform a series of exercise activities; capturing, via the fitness testing unit, a baseline heart rate, a maximum heart rate, and a recovery heart rate of the individual while the individual is performing the series of exercise activities; and determining a fitness level of the individual using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate, wherein determining the fitness level of the individual using the age of the individual employs a factor indicating highest potential heart rate recovery given the age of the individual.
2. The method of claim 1 wherein the series of exercise activities includes exercises of varying intensity.
3. The method of claim 2 wherein the series of exercise activities includes: a relaxation activity, a cardiovascular activity, and a recovery activity.
4. The method of claim 3 wherein the baseline heart rate is captured while the individual is performing the relaxation activity, the maximum heart rate is captured while the individual is performing the cardiovascular activity, and the recovery heart rate is captured while the individual is performing the recovery activity.
5. The method of claim 3 wherein capturing includes continuously monitoring heart rate of the individual and instructing the individual to perform the series of activities includes: instructing the individual, via the GUI, to perform the cardiovascular activity until it is determined that the monitored heart rate is at 80% of a recommended heart rate, wherein the recommended heart rate is based on the age of the individual.
6. The method of claim 1 further comprising: obtaining the age of the individual by accessing a database storing a user profile associated with the individual.
7. The method of claim 1 wherein the fitness testing unit comprises a bicycle or a treadmill.
8. The method of claim 1 further comprising, at least one of: storing the determined fitness level; displaying the determined fitness level in the GUI; aggregating the determined fitness level of the individual with fitness level data for multiple individuals; and providing exercise instructions to the individual based upon the determined fitness level.
9. A fitness testing system for determining a fitness level of an individual, the system comprising: a processor and a memory with computer code instructions stored thereon, the processor and the memory, with the computer code instructions, being configured to cause the fitness testing system to: instruct an individual, via a graphical user interface (GUI) of the fitness testing system, to

perform a series of exercise activities; capture a baseline heart rate, a maximum heart rate, and a recovery heart rate of the individual while the individual is performing the series of exercise activities; and determine a fitness level of the individual using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate, wherein determining the fitness level of the individual using the age of the individual employs a factor indicating highest potential heart rate recovery given the age of the individual.

10. The system of claim 9 wherein the series of exercise activities includes exercises of varying intensity.

11. The system of claim 10 wherein the series of exercise activities includes: a relaxation activity, a cardiovascular activity, and a recovery activity.

12. The system of claim 11 where, in capturing the baseline heart rate, the maximum heart rate, and the recovery heart rate, the processor and the memory, with the computer code instructions, are configured to cause the fitness testing system to: capture the baseline heart rate while the individual is performing the relaxation activity; capture the maximum heart rate while the individual is performing the cardiovascular activity; and capture the recovery heart rate while the individual is performing the recovery activity.

13. The system of claim 11 where, in the capturing and instructing, the processor and the memory, with the computer code instructions, are further configured to cause the fitness testing system to: continuously monitor heart rate of the individual; and instruct the individual, via the GUI, to perform the cardiovascular activity until it is determined that the monitored heart rate is at 80% of a recommended heart rate, wherein the recommended heart rate is based on the age of the individual.

14. The system of claim 9 wherein the processor and the memory, with the computer code instructions, are further configured to cause the fitness testing system to: obtain the age of the individual by accessing a database storing a user profile associated with the individual.

15. The system of claim 9 wherein the fitness testing system is integrated into a bicycle or treadmill.

16. The system of claim 9 wherein the processor and the memory, with the computer code instructions, are further configured to cause the fitness testing system to perform at least one of: storing the determined fitness level; displaying the determined fitness level in the GUI; aggregating the determined fitness level of the individual with fitness level data for multiple individuals; and providing exercise instructions to the individual based upon the determined fitness level.

17. A computer program product for determining a fitness level of an individual using a fitness testing unit, the computer program product comprising: one or more non-transitory computer-readable storage devices and program instructions stored on at least one of the one or more storage devices, the program instructions, when loaded and executed by a processor associated with the fitness testing unit, cause the fitness testing unit to: instruct an individual, via a graphical user interface (GUI) of the fitness testing unit, to perform a series of exercise activities; capture a baseline heart rate, a maximum heart rate, and a recovery heart rate of the individual while the individual is performing the series of exercise activities; and determine a fitness level of the individual using an age of the individual and the captured baseline heart rate, maximum heart rate, and recovery heart rate, wherein determining the fitness level of the individual using the age of the individual employs a factor indicating highest potential heart rate recovery given the age of the individual.

18. The computer program product of claim 17 wherein the series of exercise activities includes: a relaxation activity, a cardiovascular activity, and a recovery activity.

19. The method of claim 18 where, in capturing the baseline heart rate, the maximum heart rate, and the recovery heart rate, the program instructions cause the fitness testing unit to: capture the baseline heart rate while the individual is performing the relaxation activity; capture the maximum heart rate while the individual is performing the cardiovascular activity; and capture the recovery heart rate while the individual is performing the recovery activity.

20. The computer program product of claim 17 wherein, the program instructions cause the fitness testing unit to perform at least one of: storing the determined fitness level; displaying the determined fitness level in the GUI; aggregating the determined fitness level of the individual with fitness level data for multiple individuals; and providing exercise instructions to the individual based upon the determined fitness level.

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